Projects
Inbox 7
Model
Rules
Benchmark
Revisions
Simulate

Simulate

Evacuation as a design-review tool · people are neutral dots · results come from runs, never from hand

Scenario

Rev B · 14:32

OCCUPANCY

Day **Night**

Night · 68 people start in their Deck 5 cabins.

Inflow to Stair 3B · Deck 6 44 · Deck 4 40 · Deck 7 24

ROUTES

Primary **Secondary**

Which route is out

- ○ Stair 3A — MVZ 5
- ○ Stair 3B — Decks 4–7
- ◉ FD-5-165P — Fr 165 · P
- ○ FD-5-165S — Fr 165 · S

FIRE WHAT-IF — On

Origin

Service · Deck 5 · Fr 161 · CL

Pick on plan

Smoke spread is illustrative (zone model), not a CFD result.

RUNS

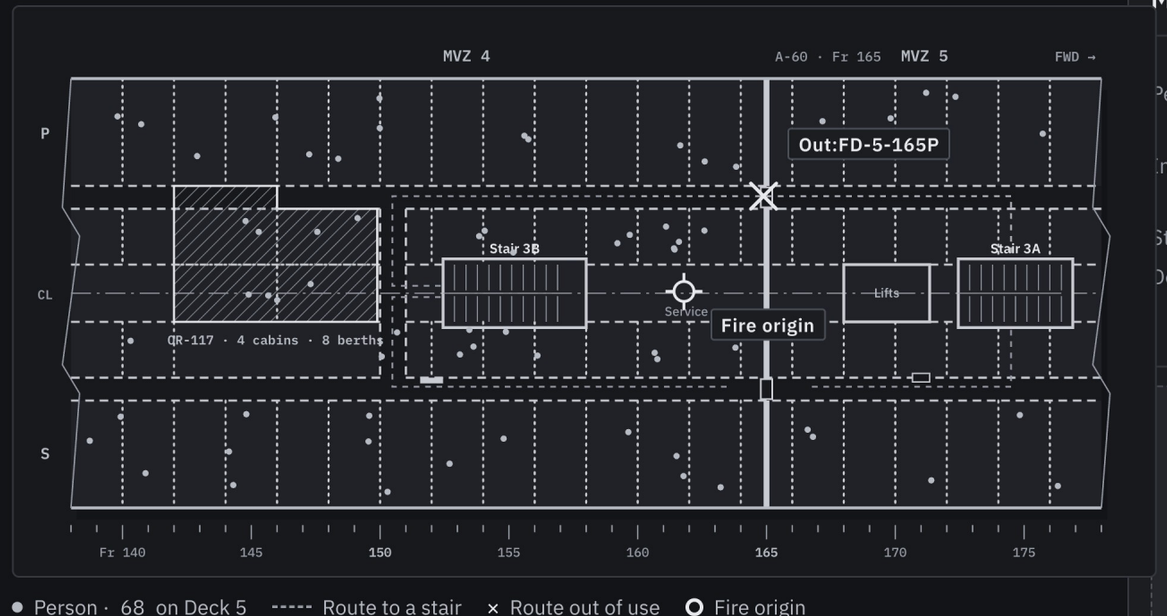
– 500 +

Each run varies walking speed and start delay.

Run 500

Deck 5 · Rev B before fix 2 · click the plan to move the fire origin

Show findings



● Person · 68 on Deck 5 ----- Route to a stair × Route out of use ○ Fire origin

Status colours stay with findings; none are used here.

A-60 A-0 B-15 B-0

Model reader

From the model

People	68 on Deck 5 · berths in IfcSpace
Inflow	Stair 3B · D6 44 · D4 40 · D7 24
Stairs	3A · 3B (clear 1,350 mm)
Doors	fire rating from Pset_DoorCommon

2 doors have no fire rating · treated as open

D-5-171-S (U-01) and 1 more on Deck 5. Smoke passes through them in the what-if.

Resolve before running

SCENARIO IN ONE SENTENCE

Night on Deck 5, Rev B: 68 people head for Stairs 3A and 3B while 108 more arrive at Stair 3B from Decks 6, 4 and 7, with MVZ door FD-5-165P out of use and a fire starting in the service space at Fr 161, over 500 runs.

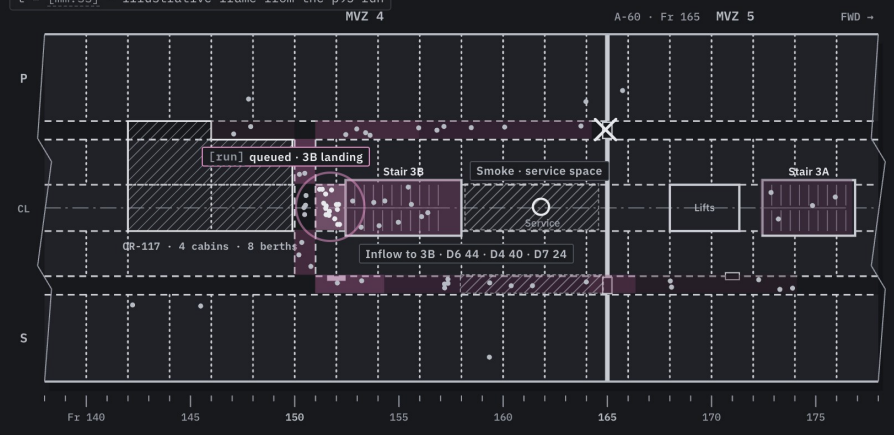
← Scenario

Night evacuation · Deck 5

Night · Secondary · FD-5-165P out · Fire what-if · Fr 161 · CL · 500 runs

finished [run]

t = [mm:ss] · illustrative frame from the p95 run



Density · heatmap at 40%
<0.5 0.5–1 1–2 2–3.5 >3.5 p/m²

Smoke (illustrative)
Person
Person waiting

A-60
A-0
B-15
B-0

[mm:ss] / [mm:ss]

x1 x10 · 2D 3D

Smoke reaches corridor · [run] · Queue peak · [run] · Last person out · [run]

P95 EVACUATION TIME

Simulation · 500 runs

[run] mm:ss



QUEUE AT STAIR 3B LANDING

persons · time



CONGESTION · SORTED AFTER RUN

peak p/m²

Stair 3B landing · Deck 5		[run]
Cross corridor · Fr 150–151		[run]
Deck 5 · Fr 152 · S	F-09 pinch	[run]
FD-5-165S · Fr 165 · S	MVZ door	[run]

✓ AI explanation · numbers checked

Stair 3B takes the 68 people on Deck 5 and 108 more from Decks 6, 4 and 7 (44 + 40 + 24). With FD-5-165P out, port-side people aft of Fr 165 reach Stair 3A only through the starboard door, so most join the 3B queue. It peaks at [run] people at [run].

The corridor pinch at Fr 152 · S (850 mm, F-09) is next at [run] p/m².

Pin to review · Try fixes

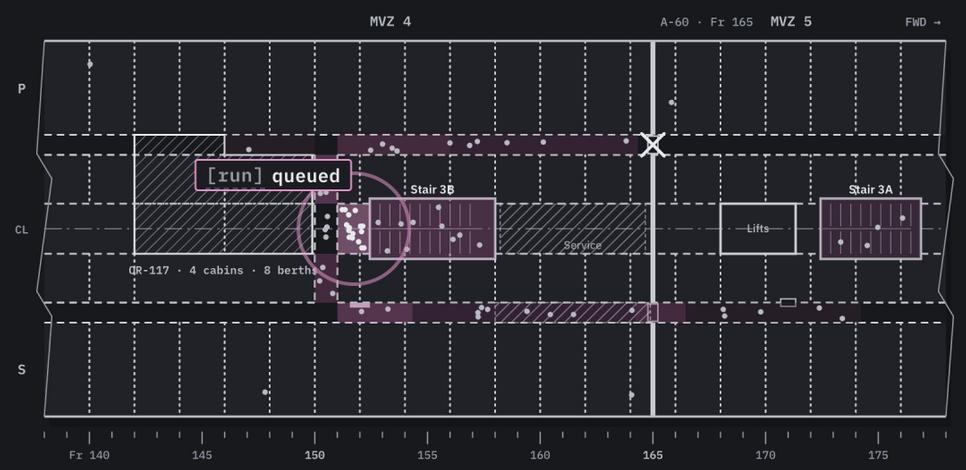
Deck 5 sub-model with inflows from Decks 4, 6, 7 · design insight, not a full-ship Circ.1533 analysis

← Results

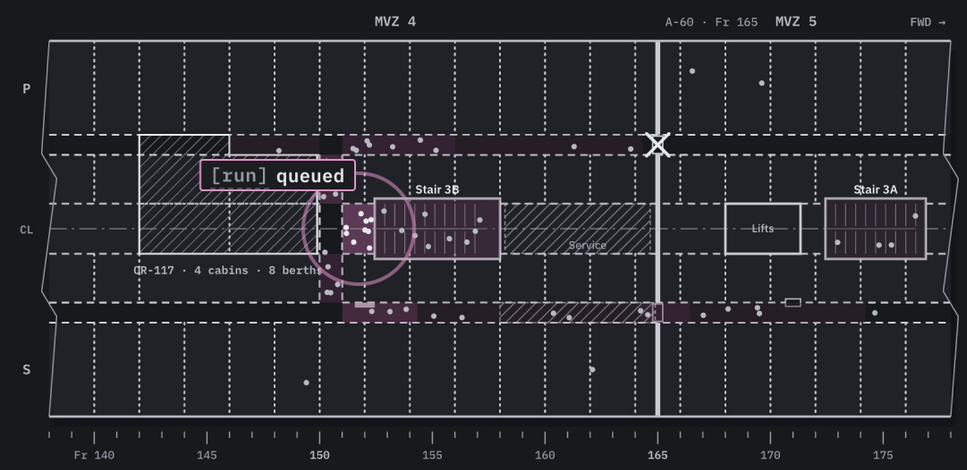
Compare · before and after a fix

Night · Secondary · FD-5-165P out · Fire what-if · Fr 161 · CL · 500 runs each

Rev B 68 on Deck 5 · before fix — t = [mm:ss]



Rev B + fix 2 64 on Deck 5 · N₁ 64 — t = [mm:ss]



▶ Synced — [mm:ss] x1 x10

P95 EVACUATION TIME
[run] → [run] mm:ss
Δ [run] · Rev B → fix 2

PEAK DENSITY
[run] → [run] p/m²
Δ [run] · Rev B → fix 2

QUEUE PEAK · 3B LANDING
[run] → [run] people
Δ [run] · Rev B → fix 2

1 · Widen Stair 3B to 1,400 mm — Pass

1,380 ≤ 1,400 mm · margin 20 mm

Rule engine new p95 [run]

Needs Structure review · deck openings on Decks 4–7.

Replay on the right

2 · Keep four cabins, one berth each — Pass

N₁ 64 → 1,340 ≤ 1,350 mm · margin 10 mm

Rule engine new p95 [run]

Applied in FreeCAD at 14:36 · Rev B.

Shown on the right

3 · One cabin instead of four — Pass

N₁ 62 → 1,320 ≤ 1,350 mm · margin 30 mm

Rule engine new p95 [run]

Changes the scope of OR-117.

Replay on the right

▸ 1 idea rejected by the engine · not simulated

SonarIQ Demo ship · Rev B Live

Profile: Passenger ship · > 36 passengers

Scope: Selection | **Deck 5** | Whole model

Packs: ☑ Fire & life safety ☑ Accessibility ☑ Class · DNV To verify

Analyse 14 re-run · 0.4 s · Rev B · 14:32

4	1	1	86	4
Fail	Warning	Cannot verify	Pass	Dormant

Findings 6 | Chat | History

Rev B · 14:33

YOU
simulate a night evacuation

SIMULATION PREVIEW AI draft · you decide

Scenario	Night · Primary (all routes open)
People	68 on Deck 5
Inflow 3B	D6 44 · D4 40 · D7 24
Runs	500
Model	2 doors without fire rating, treated as open

Confirm Cancel Nothing runs until you confirm.

You confirmed · 14:33

Simulating [n] of 500 runs

Rule checks are not affected. Cancel

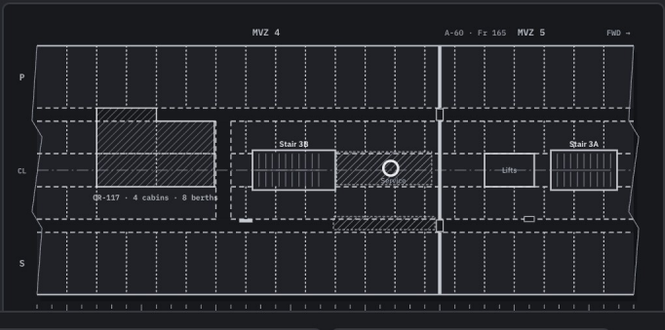
Night evacuation · Rev B Simulation · 500 runs

p95 time	[run] mm:ss
Peak density	860 px panel fold · content unrolled below
Longest route	[run] m from [run]

Open replay Re-run

YOU
what if a fire starts in the service space?

FIRE WHAT-IF · PREVIEW Illustrative smoke spread



simulate a night evacuation | what if Stair 3B is out?

Ask a question or describe a change Send